Introduction to LaTeX

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2. Your First LaTeX Document
3. Basic Language Features
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5. Specialized Applications
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Figure: Donald Knuth in 2005. *Source: Wikimedia Commons. This file is licensed under the Creative Commons Attribution-ShareAlike 2.5 Generic license.*

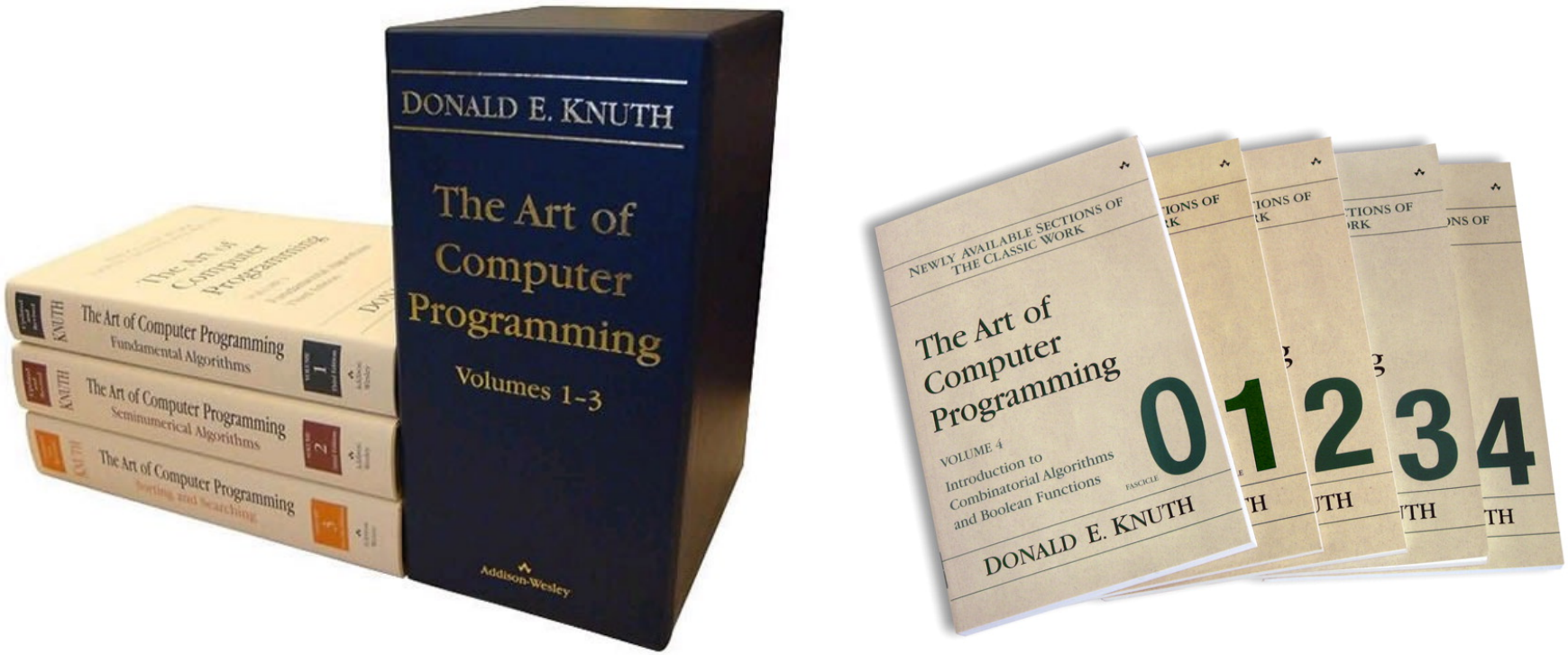


Figure: *The Art of Computer Programming. Source: MSDN, Pearson Education, Informit. All rights reserved.*



Figure: The TeX logo.

```
\newif\ifFPD@overflow
\newdimen\FBD@denom
\def\fpdivide#1#2{%
  \FPD@overflowfalse
  \ifdim\AbsValD#2<1\p@
    \begingroup\FBD@denom\ifdim#2<\z@-\fi5000#2%
    \let\next\@empty
    \ifdim\AbsValD#1>\FBD@denom
      \def\next{%
        \FPD@overflowtrue
        \debug2{Overflow dividing \the#1 by \the#2 -> inf}%
        #1=5000\p@}%
    \fi
    \ifdim\AbsValD#2<.001\p@\ifdim\AbsValD#2<.001\p@
      \def\next{%
        \FPD@overflowtrue
        \debug2{Overflow dividing \the#1 by \the#2 -> 0}%
        #1=0\p@}%
    \fi\fi
    \expandafter\endgroup\next
  \fi
```

Figure: Some TeX code. *Source: The Lone TeXnician. All rights reserved.*

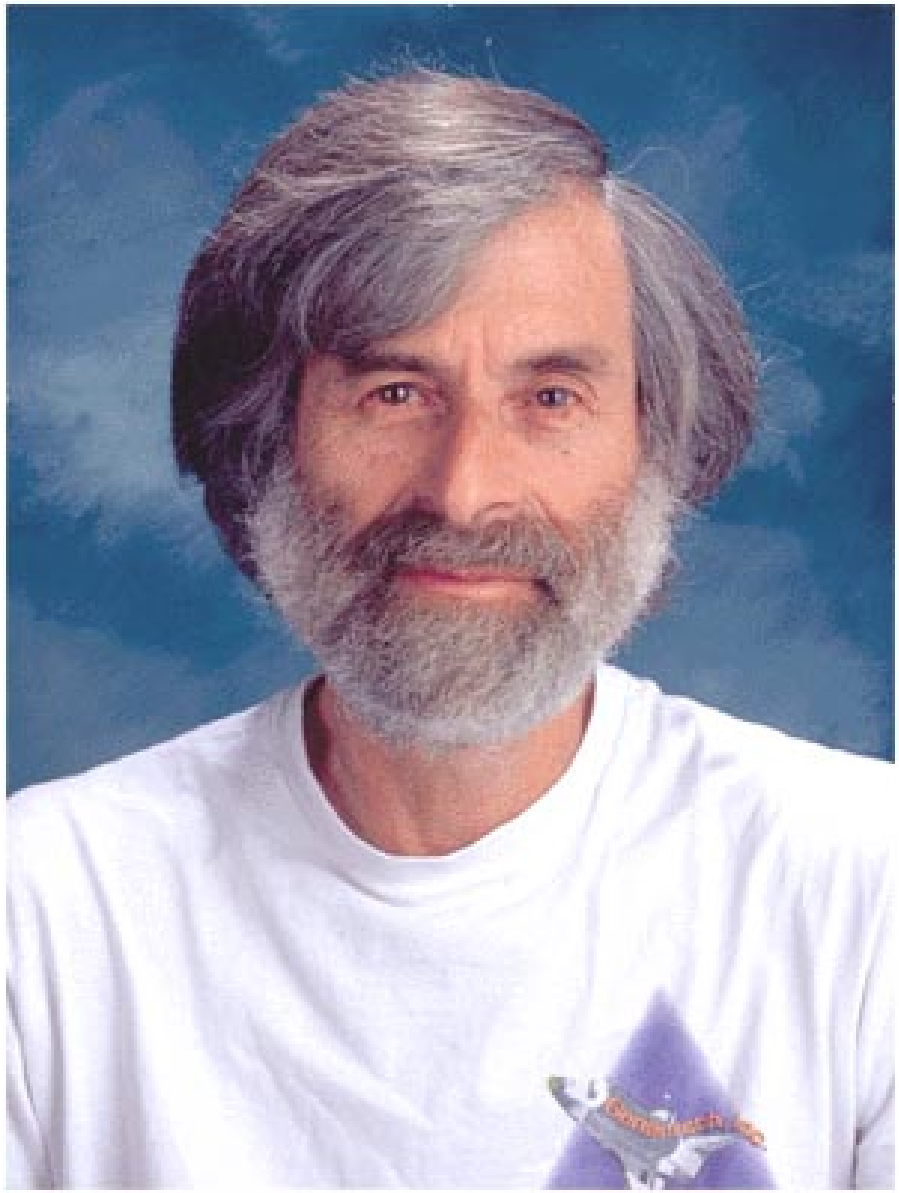


Figure: Leslie Lamport in 2004. *Source: Wikimedia Commons. Licensed for any purpose.*

```
\section{Problem 4}
\begin{enumerate}
  \setcounter{enumi}{24}
\item
  \begin{enumerate}
  \item $W$ is the interior volume of a half-paraboloid
    opening in the $+z$ direction, truncated on the top by
    $z = 4 - y^2$, which looks a bit like the roof of a
    greenhouse, and truncated on the bottom by the $xy$
    plane. The paraboloid has equation $z = x^2 + 3y^2$,
    and its shadow in the $xy$ plane is $x^2/4 + y^2 \le 1$.

  \item The appropriate integral is
    \begin{align}
      \int_0^1 \int_{-2\sqrt{1-y^2}}^{2\sqrt{1-y^2}}
      \int_{x^2+3y^2}^{4-y^2} \left(x^3 + y^3\right)dzdxdy.
    \end{align}
```

Figure: Some LaTeX code.



Figure: The LaTeX logo.

What is LaTeX?

LaTeX is a sophisticated document preparation system and desktop publishing utility.

LaTeX has ...

- Footnotes and endnotes
- Bibliography support
- Reference tracking
- Stylistic uniformity
- Crazy algorithms

However ...

- *LaTeX is not a word processor!*

What is LaTeX not?

LaTeXis a programming language, not a word processor.

LaTeX will *not* . . .

- Spell-check your documents
- Give you complete control over the way your document looks
- Let you see your document while you are writing it

Core LaTeX philosophy:

You take care of writing; we'll take care of presentation.

- Humans write text.
- Computers figure out how to display the text.

Why should I use LaTeX?

Sometimes, presentation gets in the way of content.

Example: underlining vs. italics

- Word processor way: set italics and/or underlining each time
- LaTeX way: tell LaTeX to *emphasize*; set what that means later

Example: journal article / thesis

- Word processor way: risk accidentally modifying provided template
- LaTeX way: write your text, let LaTeX worry about layout

Why should I not use LaTeX?

- Generally slower (exception: mathematics)
- Lack of complete control

Your first LaTeX document

4 basic steps

1. Write a .tex file using your favorite text editor
2. Typeset using LaTeX or PDFLaTeX
3. Preview the result using xdvi or xpdf (or Acrobat Reader or Evince)
4. (optional) Convert the result to PostScript and print

1. Write a .tex file

hello.tex

```
\documentclass{article}
\begin{document}
Hello, world!
\end{document}
```

2. Typeset using LaTeX

In a terminal:

```
$ cd path/to/folder/containing/your/.tex/file
$ latex hello.tex
```

2. Typeset using LaTeX

Result:

```
This is pdfTeX, Version 3.1415926-1.40.10 (TeX Live 2009)
entering extended mode
(./hello.tex
LaTeX2e <2009/09/24>
Babel <v3.8l> and hyphenation patterns for english, usenglishmax, dumylang, noh
yphenation, german-x-2009-06-19, ngerman-x-2009-06-19, ancientgreek, ibycus, ar
abic, basque, bulgarian, catalan, pinyin, coptic, croatian, czech, danish, dutc
h, esperanto, estonian, farsi, finnish, french, galician, german, ngerman, mono
greek, greek, hungarian, icelandic, indonesian, interlingua, irish, italian, ku
rmanji, latin, latvian, lithuanian, mongolian, mongolian2a, bokmal, nynorsk, po
lish, portuguese, romanian, russian, sanskrit, serbian, slovak, slovenian, span
ish, swedish, turkish, ukenglish, ukrainian, uppersorbian, welsh, loaded.
(/usr/local/texlive/2009/texmf-dist/tex/latex/base/article.cls
Document Class:  article 2007/10/19 v1.4h Standard LaTeX document class
(/usr/local/texlive/2009/texmf-dist/tex/latex/base/size10.clo))
No file hello.aux.
(./hello.aux) )
Output written on hello.dvi (1 page, 232 bytes).

Transcript written on hello.log.
```

3. Preview using xdvi

New files!

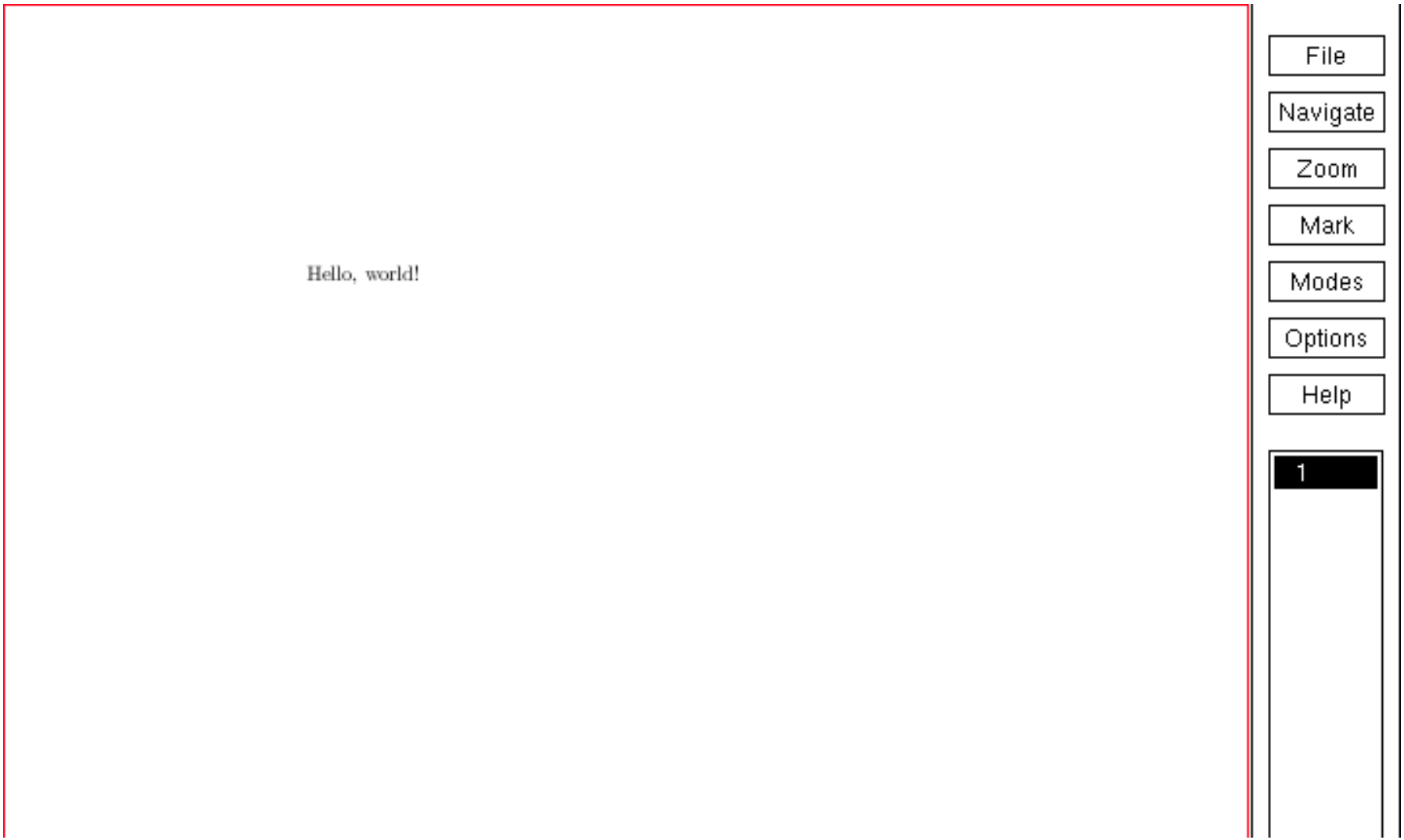
- hello.aux
- hello.dvi
- hello.log

hello.dvi

```
$ xdvi hello.dvi
```

3. Preview using xdvi

Result:



4. Convert to PostScript and print

Converting to PostScript

```
$ dvips -t letter -o hello.ps hello.dvi
```

Printing

Don't run this command right now!

```
$ lpr -Pprintername hello.ps
```

When LaTeX complains

Overfull/underfull hbox

LaTeX couldn't make your text fit nicely on one line.

Overfull/underfull vbox

LaTeX couldn't make your text fit nicely on a page.

Runaway argument

You forgot to close a brace.

Solution

1. Type x and hit enter
2. Fix the error
3. Re-run LaTeX

Sample document 1

"Synthesizing Congestion Control Using Replicated Archetypes"
Generated by SCIgen, the automatic computer science paper generator
`http://pdos.csail.mit.edu/scigen/`

Declarations and environments

Declarations . . .

- Are stated once
- Take effect until further notice
- Can be constrained using curly braces

Example: \documentclass

Environments . . .

- Have corresponding \begin and \end declarations
- Apply formatting to their contents

Example: \begin{document} / \end{document}

The \documentclass declaration

\documentclass tells LaTeX what basic document template to use.

Other templates ("classes"):

- book
- report
- letter
- revtex4-1
- thesis
- beamer

Sectioning declarations

- \part (book only)
- \chapter (book and report only)
- \section
- \subsection
- \subsubsection
- \paragraph
- \subparagraph
- \subsubparagraph

Example: \chapter{A Mad Tea-Party}

Arguments

Arguments can be required or optional.

Required arguments . . .

- Are placed in curly braces
- Cause LaTeX to complain if left out

Example: \documentclass{**article**}

Optional arguments . . .

- Are placed in square brackets
- Don't cause errors if left out
- Come before required arguments

Example: \documentclass[**12pt,letterpaper**]{article}

The title

Place in preamble (before \begin{document}):

```
\title{Synthesizing Congestion Control Using Replicated Archetypes}
\author{Benjamin Barenblat\\MIT \and SCIgen\\CSAIL}
\date{\today}
```

Place in document:

```
\maketitle
```

Some classes allow for more preamble commands.

Including graphics

Straight LaTeX requires documents to be in encapsulated PostScript format.

Place in preamble:

```
\usepackage{graphicx}
```

Place in document:

```
\begin{figure}
  \begin{center}
    \includegraphics{doc1/flowchart.eps}
  \end{center}
  \caption{The diagram used by Oxymel.}
\end{figure}
```

Labeling figures

Place after caption:

```
\label{robots}
```

Place in appropriate location:

```
... figure~\ref{robots}
```

You will have to run LaTeX twice!

Labeling figures and stuff

Place after appropriate command:

```
\label{robots}
```

Place in appropriate location:

```
... \ref{robots}
```

You will have to run LaTeX twice!

Tables

Recall figures:

```
\begin{figure}
  \begin{center}
    \includegraphics{doc1/flowchart.eps}
  \end{center}
  \caption{The diagram used by Oxymel.}
\end{figure}
```

Similar method for tables:

```
\begin{table}
  \begin{center}
    \includegraphics{doc1/datatable.eps}
  \end{center}
  \caption{Our raw data.}
\end{table}
```

Tabular

Code:

```
\begin{tabular}{ℓ ℓ ℓ}
  Language & Seek time & Write time\\
  \hline
  BLooP & 27 & 42\\
  FLooP & 12 & 19\\
  GLooP & 11 & 22
\end{tabular}
```

Result:

Language	Seek time	Write time
BLooP	27	42
FLooP	12	19
GLooP	11	22

Lists

Lists can be numbered (enumerated) or bulleted (itemized).

Numbered lists:

```
\begin{enumerate}
   \item Item 1
   \item Item 2
\end{enumerate}
```

Bulleted lists:

```
\begin{itemize}
   \item Item 1
   \item Item 2
\end{itemize}
```

Quoting other works

quote

```
\begin{quote}
  Here's a single-paragraph quote.
\end{quote}
```

quotation

```
\begin{quotation}
  Here's a multiparagraph quote.
  Here's the second paragraph.
\end{quotation}
```

verse

```
\begin{verse}
  Here's some poetry.\\
  Here's the second line.\\
\end{verse}
```

Finishing touches

The abstract:

```
\begin{abstract}
   Yada yada yada....
\end{abstract}
```

A title page

```
\documentclass[titlepage]{article}
```

Real headers

```
\pagestyle{headings}
```

Miscellaneous

Spaces

~	nonbreaking space
\	force normal interword space
	(e.g., `Steele et al.\ discovered`)
\@.	force end-of-sentence space
	(e.g., `I program in C\@.  You?`)
\hspace{1in}	make horizontal space
\vspace{1in}	make vertical space

Breaking

\\	force new line
\newpage	force new page
\noindent	force no indentation of current paragraph

Comments: Anything after % on a single line is ignored.

Customizing LaTeX

Some customization commands are built-in.

Changing font face:

\emph{*text*}, \textnormal{text}, \textrm{text}, \textsf{text}, \texttt{text}, \textbf{**text**}, \textit{*text*}, \textsc{TEXT}

Changing font size:

\tiny, \scriptsize, \footnotesize, \small, \normalsize, \large, \Large, \LARGE, \huge, \Huge

Changing alignment:

\begin{center}, \begin{flushright}, \begin{flushleft}

Customizing LaTeX

Customizations can also occur through *packages*.

Including a package:

\usepackage{packagename}

Useful packages

graphicx, geometry, setspace, fancyhdr, calc, mathpazo, microtype, amsmath, amsfonts, amsthm, amssymb, url, ulem, textcomp, listings, eco, mathtools, mhchem, units, wrapfig, color, ccaption, titlesec, epstopdf, tabularx, tocloft ...

A survey of useful packages

geometry

Controls margins:

```
\usepackage[margin=1.1in]{geometry}
```

setspace

Allows you to use double and 1.5 spacing:

```
\usepackage{setspace}
\doublespacing
```

fancyhdr

Controls header and footer:

```
\usepackage{fancyhdr}
\pagestyle{fancy}
\fancyhf{} % Reset header and footer
\fancyhead[R]{\thepage} % This puts the page in the right of the header
```

Changing fonts

Fonts are usually loaded through packages as well.

\usepackage[urw-garamond]{mathdesign}	Garamond
\usepackage{mathpazo}	Palatino
\usepackage[scaled]{helvet}	Helvetica
\usepackage{courier}	Courier
\renewcommand*\sfdefault{uop}	Optima
\usepackage{concrete}	Computer Concrete
\usepackage{tgbonum}	Bookman
\usepackage{txfonts}	Times

More fonts are available at The LaTeX Font Catalogue, `http://www.tug.dk/FontCatalogue/`.

Typesetting mathematics

LaTeX's math support far outstrips that of any other available piece of software.

The Leibniz integral rule

$$\frac{d}{d\alpha}\int_{a(\alpha)}^{b(\alpha)} f(x,\alpha)dx = \frac{db(\alpha)}{d\alpha}f\big(b(\alpha),\alpha\big) - \frac{da(\alpha)}{d\alpha}f\big(a(\alpha),\alpha\big) + \int_{a(\alpha)}^{b(\alpha)} \frac{\partial}{\partial\alpha}f(x,\alpha)dx$$

Generalized Stokes theorem

If ω is an $(n-1)$-form with compact support on M and ∂M denotes the boundary of M with its induced orientation, then

$$\int_M \mathrm{d}\omega = \oint_{\partial M} \omega.$$

Text and math modes

LaTeX is always operating in either text mode, display math mode, or inline math mode.

Inline math mode

- Enter/exit using \$...\$ or \(...\)
- Large symbols and super/subscripts are squashed:

$$\int_1^\infty e^- x dx \qquad \sum_{n=0}^\infty n! \qquad \left[\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right]$$

Display math mode

- Enter/exit using \begin{equation}...\end{equation} or \[...\]
- Large symbols and super/subscripts are displayed in full glory
- Equations can be numbered

$$\int_1^\infty e^- x dx \qquad \sum_{n=0}^\infty n! \qquad \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Basic mathematics

The vast majority of math commands are highly logical.

974	974	x	x
$4+2$	4 + 2	49 ± 71	49 \pm 71
$\sqrt[3]{5}$	\sqrt[3]{5}	$\phi \in U$	\phi\in U
x_1^2	x_1^2	$f''(\xi)$	f''(\xi)
$\frac{x}{y}$	\frac{x}{y}	$\forall x \exists y$	\forall x\exists y
$\sum_{k=1}^{n} k$	\sum_{k=1}^n k	$U \cap V$	U\cap V
$x \leqslant y$	x \leqslant y	$P \Leftrightarrow Q$	P\Leftrightarrow Q
$2 \neq 4$	2 \neq 4	$\mathbb{R} \subset \mathbb{C}$	\mathbb{R}\subset \mathbb{C}
$\nabla \cdot \mathbf{\Psi}$	\nabla\cdot\mathbf{\Psi}		
$\hat{\imath} \times \hat{\jmath} = \hat{k}$	\hat{\imath}\times\hat{\jmath}=\hat{k}		

Detexify[2] (`http://detexify.kirelabs.org/`) gives commands for any symbol.

Mathematics packages and environments

- Use \usepackage{amsfonts,amsmath,amssymb,amsthm} unless you have a good reason not to.
- \usepackage{esint} will get you cool integral signs.

equation

$$\oiint_{\partial\Omega} \mathbf{F} \cdot d\mathbf{S} = \iiint_{\Omega} \nabla \cdot \mathbf{F} dxdydz \tag{1}$$

equation*

$$\oiint_{\partial\Omega} \mathbf{F} \cdot d\mathbf{S} = \iiint_{\Omega} \nabla \cdot \mathbf{F} dxdydz$$

The Short Math Guide for LaTeX (`http://www.ams.org/tex/short-math-guide.html`) has a full listing.

Mathematics packages and environments

align

```
\begin{block}
  \begin{align}
    a &= \oiint_{\partial\Omega}\mathbf{F}\cdot
        d\mathbf{S}  &  (3)(5+7) &= (3)(12)\\
      &= \iiint_\Omega\nabla\cdot\mathbf{F}dxdydz
      &  &= 36
  \end{align}
\end{block}
```

$$a = \oiint_{\partial\Omega} \mathbf{F}\cdot d\mathbf{S} \qquad (3)(5+7) = (3)(12) \qquad (2)$$

$$= \iiint_{\Omega} \nabla\cdot\mathbf{F}dxdydz \qquad = 36 \qquad (3)$$

Labeling figures and stuff

Place after appropriate command:

```
\label{robots}
```

Place in appropriate location:

```
... \ref{robots}
```

You will have to run LaTeX twice!

Labeling figures and equations and stuff . . .

Place in environment:

```
\label{gaussthm}
```

Place in appropriate location:

```
. . . equation \ref{gaussthm}
```

You will have to run LaTeX twice!

Your thesis



"Piled Higher and Deeper" by George Cham http://www.phdcomics.com/

Your thesis

1. Get the templates.
2. Modify the templates to generate your skeleton.
3. Write your thesis.
4. Typeset the thesis.

Your thesis

1. Get the templates.
2. Modify the templates to generate your skeleton.
3. Write your thesis.
4. Typeset the thesis.
5. Fix the errors.
6. Typeset it again.

Your thesis

1. Get the templates.
2. Modify the templates to generate your skeleton.
3. Write your thesis.
4. Typeset the thesis.
5. Fix the errors.
6. Typeset it again.
7. Print on the thesis printer.
8. Get it bound.

Your thesis

1. Get the templates.
2. Modify the templates to generate your skeleton.
3. Write your thesis.
4. Typeset the thesis.
5. Fix the errors.
6. Typeset it again.
7. Print on the thesis printer.
8. Get it bound.
9. ...

1. Get the templates

Templates are available . . .

- On Athena: `/mit/thesis/tex/`
- On the Web: `http://web.mit.edu/thesis/www/tex.html`

2. Modify the templates

Files you will see:

- abstract.tex – abstract
- appa.tex – appendix A
- appb.tex – appendix B
- biblio.tex – references
- chap1.tex – chapter 1
- contents.tex - table of contents
- cover.tex – cover page
- main.tex – LaTeX this to make your thesis
- propcover.tex – thesis proposal cover (PhD candidates only)

The basic idea is to have one file per chapter. \include{file.tex} inserts file.tex into the current document.

2. Modify the templates

Modify these files initially:

1. cover.tex – your personal information
2. main.tex – \include your chapters

3. Write your thesis

To write a new chapter:

1. Create a new file chap*n*.tex.
2. Add \include{chap*n*.tex} to the relevant location in main.tex.

4. Typeset the thesis

To generate your thesis:

Run `latex main.tex` twice.

To generate your thesis with references:

Run `latex main.tex && bibtex main && latex main.tex && latex main.tex`.

References are beyond the scope of this class. For more information, look up BibTEX online.

Where to go from here

Further resources

- *The Not So Short Introduction to LaTeX 2ε*: `http://www.ctan.org/tex-archive/info/lshort/english/lshort.pdf`
- The LaTeX 2ε cheat sheet: `http://www.stdout.org/~winston/latex/latexsheet.pdf`
- *A Short Math Guide for LaTeX*: `http://www.ams.org/tex/short-math-guide.html`
- The `texdoc` command

LaTeX on your own computer

- Linux: TeX Live (use your package manager)
- Mac OS: MacTeX: `http://www.tug.org/mactex/`
- Windows: MikTeX: `http://www.miktex.org/`

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